



△ **Superbly transparent**, mixed-cut cushion sillimanite

Sillimanite

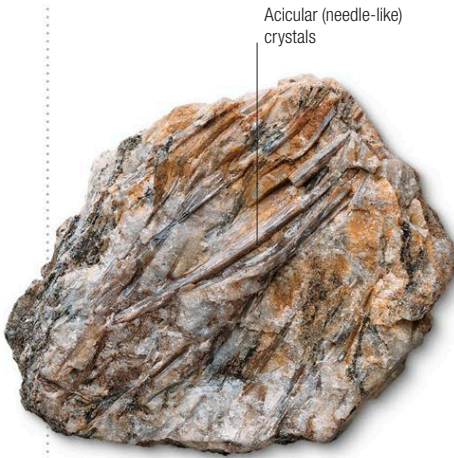
Although mainly an industrial material, transparent sillimanite is the basis of attractive faceted gemstones. Cabochons are cut from a form of sillimanite called fibrolite, so named because the mineral resembles bunches of fibres twisted together. Crystals are long, slender, and glassy, or in blocky prisms. Blue and violet are the most prized colours for gemstones. Sillimanite is distinctly pleochroic: yellowish-green, dark green, and blue can be seen within the same stone from different angles. It is a common mineral in some metamorphic rocks.

Specification

Chemical name	Aluminium silicate	Formula	Al_2OSiO_5
Colours	Colourless, blue, yellow, green, violet	Structure	Orthorhombic
Hardness	7	SG	3.2–3.3
RI	1.66–1.68	Lustre	Vitreous or silky
Streak	White	Locations	Myanmar, India, Czech Republic, Sri Lanka, Italy, Germany, Brazil, USA



Fibrous rock | Rough | This specimen of fibrous sillimanite is typical of its natural occurrence – it is rare to find gem-quality examples of the mineral.



Acicular (needle-like) crystals

Sillimanite crystals in rock | Rough | These elongated, prismatic sillimanite crystals are contained within a matrix of muscovite mica.



Faceted sillimanite | Colour variety | This Burmese gem, with a brilliant-cut crown, is a perfect example of the gem's pleochroism, showing bluish-violet and pale yellow colours.



Large oval | Cut | Found in Brazil, this stone is exceptionally large, weighing just over 21 carats, and has been faceted to enhance its yellowish-green colouring.

Mixed-cut cushion | Cut | The skilful faceting of this fibrolite sillimanite gemstone brings out its particularly fine clarity, transparency, and hue.

Pavilion facets visible through table facet



Cabochon, emerald, and scissors are the cuts most commonly used for sillimanite

Visibly fibrous cat's-eye effect



Cat's-eye cabochon | Cut | Sillimanite's own fibrous nature sometimes yields a cat's-eye effect when cut *en cabochon*, as illustrated in this stone.